

Global Vaccination Data Analysis

From Raw WHO Data to a Live, Dual-Dashboard Analytics Platform

Project Report

Covering: Data Cleaning & EDA · SQL Database Design · Cloud Migration · Power BI Dashboard · Streamlit Dashboard · GitHub Repository

GitHub Repository:

github.com/KVAIalwaysLearning/Vaccination_EDA

Streamlit App:

vaccinationeda-app0.streamlit.app

Google Drive link:

<https://drive.google.com/drive/folders/1h-TJRxzLx8gzigZuNztZg60bUM94RJ--?usp=sharing>

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1. Introduction & Project Overview

This project analyzes five WHO Immunization Data Portal extracts (1980–2023) — vaccination coverage, disease incidence rates, reported cases, vaccine introduction status, and national vaccination schedules — to understand how vaccination effort relates to disease outcomes across countries and WHO regions, and to surface gaps that matter for public health policy and resource allocation.

The project was built end-to-end as a single pipeline: raw data cleaning and exploratory analysis, a normalized relational SQL database, migration of that database to a live cloud instance, and two independent interactive dashboards — one built in Power BI and one built with Streamlit — both querying the same live database. This report documents every stage of that pipeline, with screenshots of both dashboards and references to the underlying notebooks and repository.

1.1 Pipeline at a Glance

- **Vaccination_EDA.ipynb** — Cleans, validates, and exports 7 analysis-ready CSV files from 5 raw WHO extracts.
- **SQL.ipynb** — Builds a normalized star-schema SQL database (SQLite), answers every question in the project brief with real SQL queries, generates the ER diagram, and exports Power BI-ready flattened tables.
- **Migration_n_Query.ipynb** — Migrates the entire SQLite database to a live, always-on PostgreSQL database hosted on Neon, rebuilds primary/foreign keys and indexes, and provides a password-gated interactive query tool.
- **Power BI Dashboard** — Five pages (plus a Home navigation page) connected to the live Neon database, covering KPI indicators, a geographic heatmap, trend charts, a coverage-vs-incidence scatter plot, and vaccine introduction/schedule detail.
- **Streamlit Dashboard** — A live, filterable dashboard deployed on Streamlit Community Cloud, connected directly to the same Neon database, re-querying it on every filter change.
- **GitHub Repository** — Public repository hosting every notebook, the dashboard source code, and full documentation.

2. Objectives & Data Sources

2.1 Business Objective

Analyze global vaccination data to understand trends in vaccination coverage, disease incidence, and effectiveness — clean the five source tables, reconcile them into an analysis-ready form, and use exploratory data analysis and visualization to surface patterns relevant to public health strategy, disease prevention, resource allocation, and global health policy.

2.2 Problem Statement

Identify where vaccination programs are working (rising coverage, falling incidence), where they are not (persistent low coverage, high incidence, large dose drop-off), and which vaccines/regions show introduction gaps — so that public health agencies, vaccine manufacturers, and policymakers can prioritize interventions, forecast demand, and target resources with evidence rather than guesswork.

2.3 Data Sources

Five datasets from the WHO Immunization Data Portal, covering 1980–2023:

Dataset	Rows (raw)	Grain
Vaccination coverage	399,859	Country/region/global × year × antigen × reporting category
Disease incidence rate	84,946	Country/region/global × year × disease
Reported cases	84,870	Country/region/global × year × disease
Vaccine introduction status	138,321	Country × year × vaccine name
National vaccination schedules	8,053	Country × year × vaccine × dose round

3. Data Cleaning & Exploratory Data Analysis

Notebook: `Vaccination_EDA.ipynb`

3.1 Data Cleaning Summary

Every one of the five raw files required non-trivial cleaning before it was analysis-ready:

- **Structural defect** — Every file carried a stray metadata footer row (e.g. "Created: 2025-02-01 16:00 UTC") with every real column left blank, corrupting null counts and column data types until removed.
- **Mixed aggregation levels** — The coverage table mixed genuine country-level records with pre-aggregated regional, income-group, and global rollups under a single GROUP column; country-level analysis had to filter to GROUP == 'COUNTRIES' or totals would be double-counted.
- **Invalid coverage values** — Roughly 5,100 coverage records exceeded 100% (up to 32,000%), a clear administrative-reporting artifact. These were flagged as invalid rather than silently dropped or imputed, preserving the option to include or exclude them per analysis.
- **coverage_clean built as** — Country-level, WUENIC-estimate, valid-range rows only — the subset used for every coverage comparison chart.
- **CASES nulls** — Left as true missing values rather than zero, since a real "0 cases reported" is a meaningful data point that must not be conflated with "not reported".
- **Vaccine introduction status (INTRO)** — Normalized from 9 raw values down to a clean Yes / No / Unknown flag.

3.2 Data Visualization — 15 Charts

Fifteen charts were built to explore the cleaned data, each following a why-this-chart / insight-found / business-impact structure. A quick-reference summary of each chart and its key finding, followed by the charts themselves in Section 3.2.1:

#	Chart	Key Insight
1	Global total reported cases by year (all diseases combined)	Long-run decline from 1980 through the 2000s–2010s as vaccination programs scaled up, with occasional year-specific spikes.
2	Global average WUENIC coverage over time for DTPCV3	Steady rise from the 1980s, plateauing in the 80s% range recently, with a dip aligning to pandemic-era disruption.
3	Top 10 diseases by cumulative reported cases	A small number of diseases (historically measles and pertussis) account for a disproportionate share of all cases.
4	Latest-year average DTPCV3 coverage by WHO region	Coverage is not uniform across regions — some sit well below the global plateau.
5	Vaccine introduction status (normalized) by WHO region	The share of "Yes" introduction records varies by region — some have adopted fewer newer vaccines.
6	Coverage vs incidence (measles: MCV1 vs MEASLES), latest year	Clear negative relationship — high MCV1 coverage clusters with low incidence.

7	Dose drop-off — DTPCV1 vs DTPCV3, by country, latest year	Most countries show single-digit drop-off; a meaningful subset show double-digit drop-off.
8	Coverage trend around a vaccine's national introduction year (Rotavirus example)	Coverage is near zero before the recorded introduction year and rises after, validating the introduction data.
9	Choropleth map of latest-year DTPCV3 coverage by country	Low coverage clusters geographically rather than scattering randomly, confirming regional disparities.
10	Distribution of coverage values (validity check)	The vast majority fall within the plausible 0–100% range; the >100% tail is spread out, consistent with reporting errors.
11	Cumulative count of vaccines introduced globally over time	Distinct acceleration periods follow the global availability of newer vaccines (HPV, rotavirus, PCV).
12	Bottom 10 countries by MCV1 coverage, latest year	A handful of specific countries sit far below the global plateau — a name-level priority list.
13	Measles incidence rate trend over time, by WHO region	Incidence fell across all regions since 1980, but at different rates — some regions remain persistently higher.
14	Correlation heatmap	avg_coverage correlates negatively with both total_cases and avg_incidence_rate, aggregated across all diseases.
15	Pair plot	The coverage–incidence relationship is visibly non-linear, consistent with a herd-immunity-threshold effect.

3.2.1 Charts in Full

Each chart is reproduced below with its title and the key insight found, exactly as generated in Vaccination_EDA.ipynb. Chart 9 (a choropleth map) was built as an interactive Plotly visual with no static image output, so it is not reproduced here — it is directly viewable by running the notebook.

Chart 1 — Global total reported cases by year (all diseases combined)

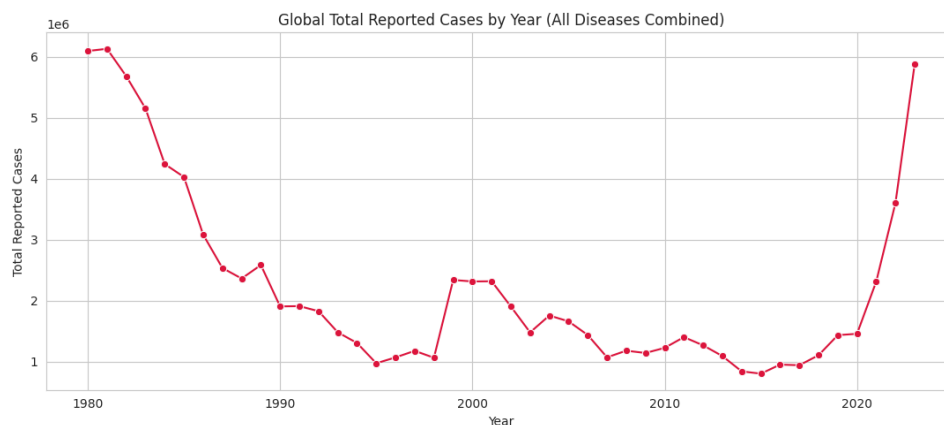


Figure 1 — Chart 1: Global total reported cases by year (all diseases combined)

Global reported cases show a long-run decline from 1980 through the 2000s–2010s as vaccination programs scaled up, though occasional spikes appear in specific years (worth cross-checking against [intro_df](#) for the diseases driving each spike).

Chart 2 — Global average WUENIC coverage over time for DTPCV3 (a core benchmark antigen)

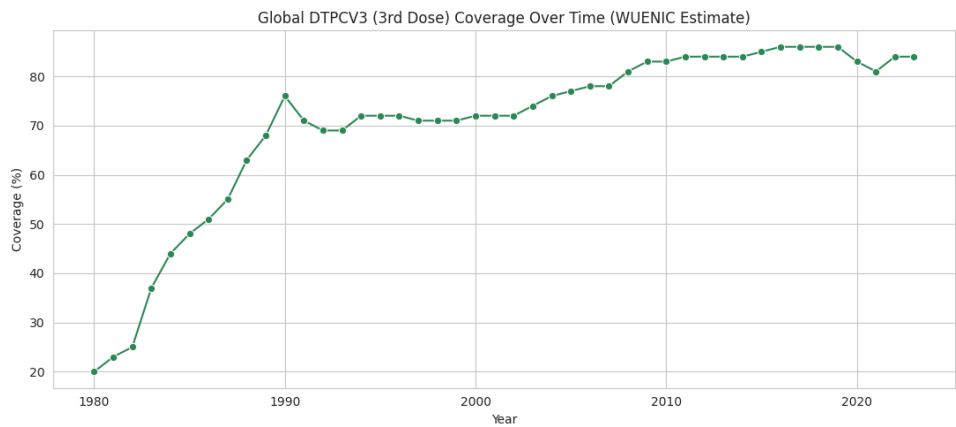


Figure 2 — Chart 2: Global average WUENIC coverage over time for DTPCV3 (a core benchmark antigen)

Global DTPCV3 coverage rose steadily from the 1980s and has plateaued in the 80s% range in recent years, with a visible dip that lines up with pandemic-era disruption to routine immunization services.

Chart 3 — Top 10 diseases by cumulative reported cases (country-level)

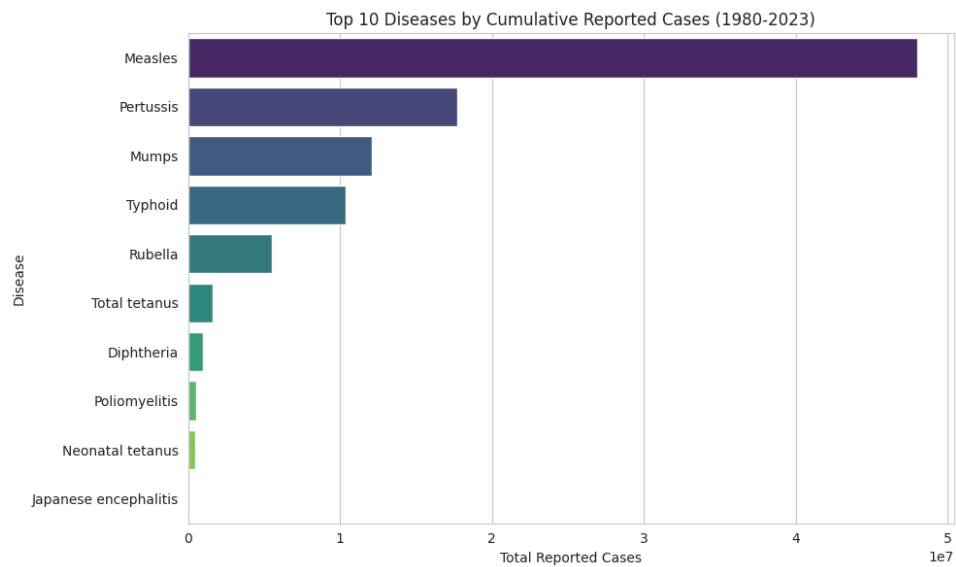


Figure 3 — Chart 3: Top 10 diseases by cumulative reported cases (country-level)

A small number of diseases (historically measles and pertussis among them) account for a disproportionate share of all reported cases across the 44-year window, making them the highest-leverage targets for continued intervention.

Chart 4 — Latest-year average DTPCV3 coverage by WHO region

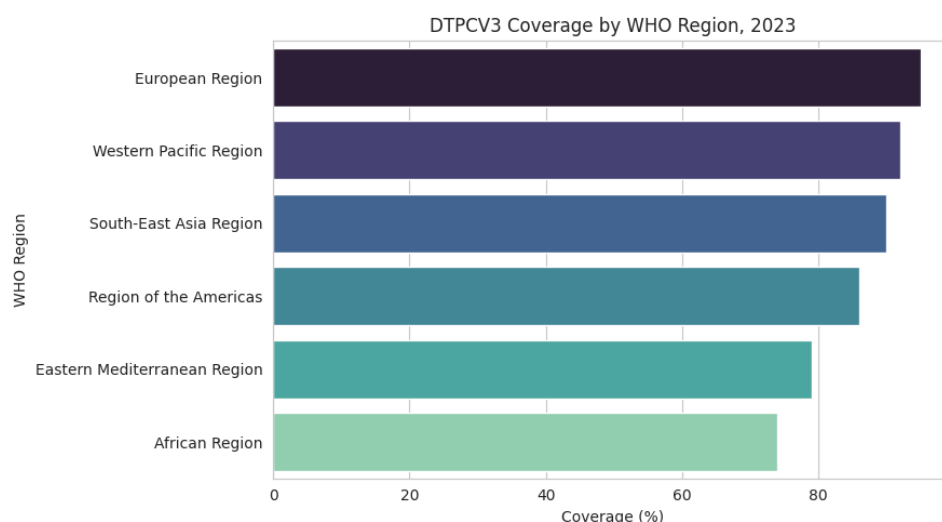


Figure 4 — Chart 4: Latest-year average DTPCV3 coverage by WHO region

Coverage is not uniform across WHO regions in the latest year — some regions sit well below the global plateau seen in Chart 2, pointing to where regional-level intervention is most needed.

Chart 5 — Vaccine introduction status (normalized) by WHO region

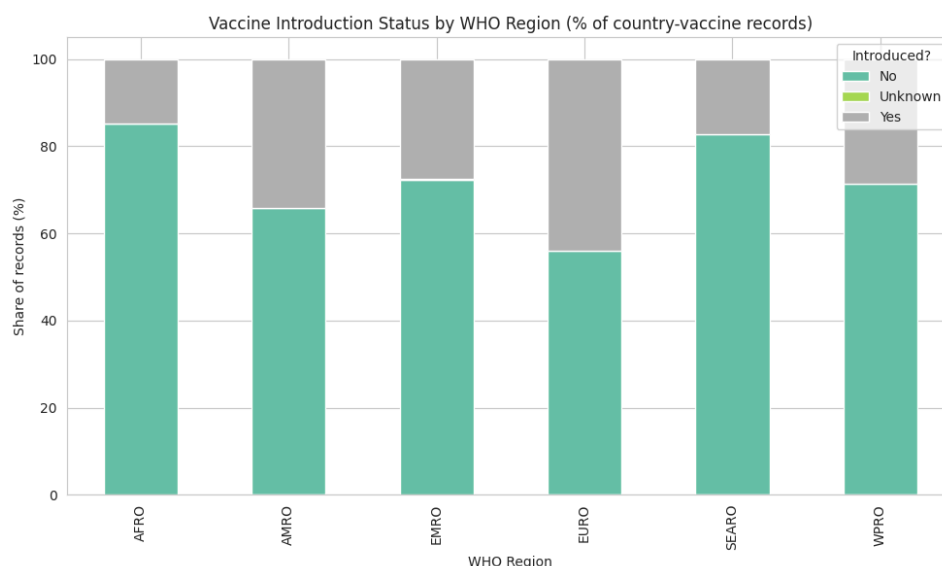


Figure 5 — Chart 5: Vaccine introduction status (normalized) by WHO region

The share of vaccine-introduction records marked "Yes" is not consistent across WHO regions — regions with a smaller Yes share have adopted fewer of the newer/available vaccines in their national programs.

Chart 6 — Country-level coverage vs incidence rate (measles: MCV1 vs MEASLES), latest common year

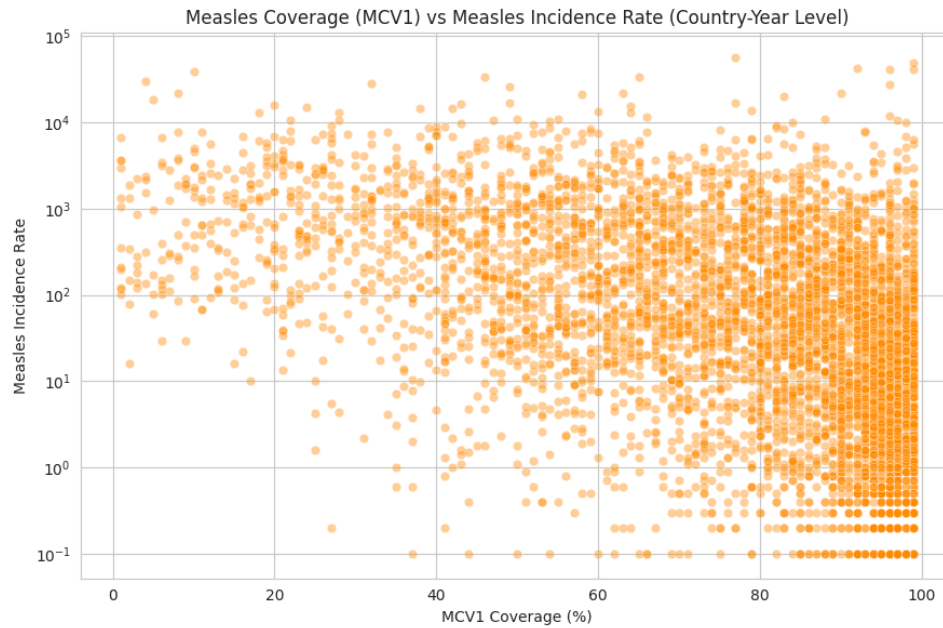


Figure 6 — Chart 6: Country-level coverage vs incidence rate (measles: MCV1 vs MEASLES), latest common year

There is a visible negative relationship between measles vaccination coverage and measles incidence rate: country-years with high MCV1 coverage cluster toward low incidence, and the printed correlation coefficient quantifies the strength of that relationship.

Chart 7 — Dose drop-off - DTPCV1 vs DTPCV3 coverage by country, latest year

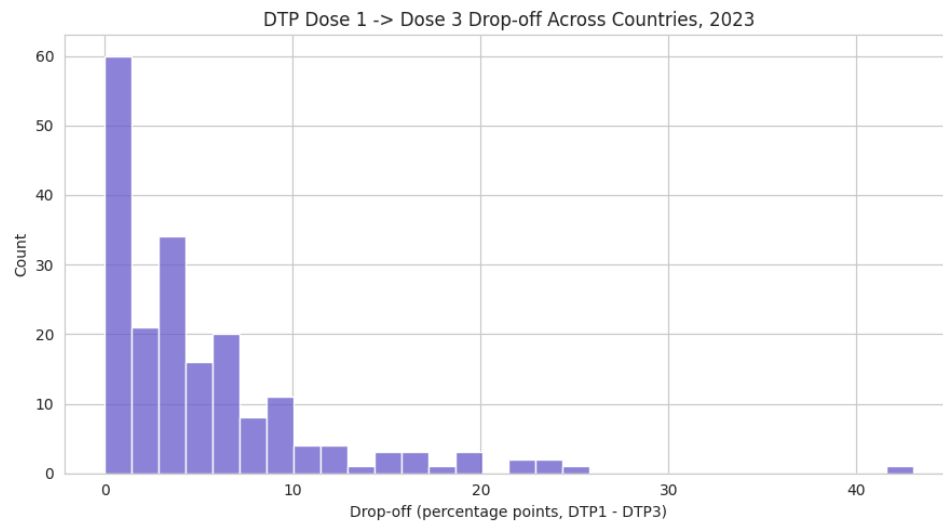


Figure 7 — Chart 7: Dose drop-off - DTPCV1 vs DTPCV3 coverage by country, latest year

Most countries show a modest single-digit drop-off between dose 1 and dose 3, but a meaningful subset show double-digit percentage-point drop-off — the printed count identifies exactly how many countries fall into that higher-risk group.

Chart 8 — Coverage trend around a vaccine's national introduction year (example: Rotavirus, sample country)

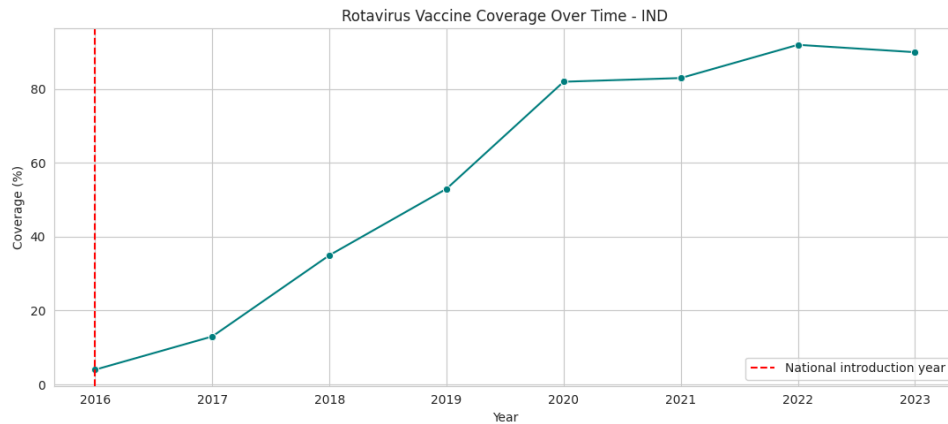


Figure 8 — Chart 8: Coverage trend around a vaccine's national introduction year (example: Rotavirus, sample country)

Coverage for the sampled vaccine/country is at or near zero before the marked introduction year and rises afterward, confirming that the introduction date recorded in `intro_df` is consistent with the coverage data in `coverage_df`.

Chart 9 — Choropleth map of latest-year DTCPV3 coverage by country

(Interactive Plotly map — no static image saved in the notebook; view live in Vaccination_EDA.ipynb.)

Low coverage is not randomly scattered — it clusters in specific geographic areas, visually confirming the regional disparities already seen in Chart 4 and pinpointing exactly which countries within those regions are weakest.

Chart 10 — Distribution of coverage values (validity check)

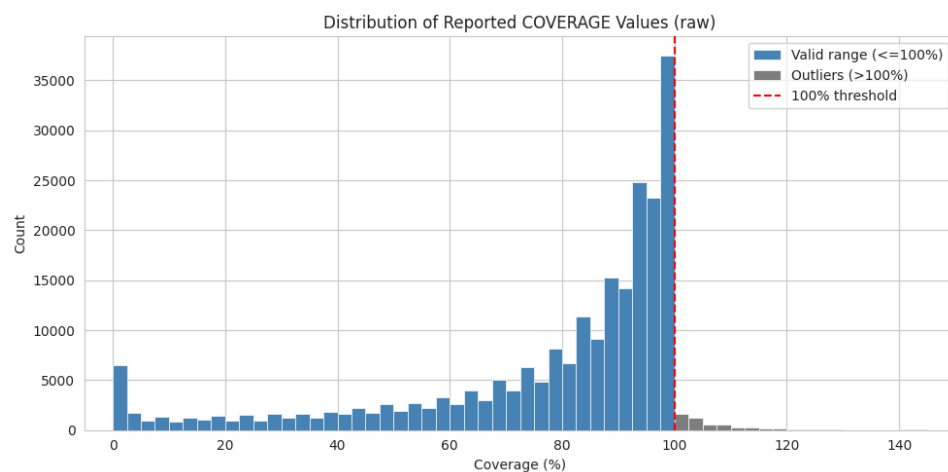


Figure 9 — Chart 10: Distribution of coverage values (validity check)

The overwhelming majority of coverage values fall within the plausible 0–100% range, and the >100% values form a distinct, spread-out tail rather than being clustered just above 100 — consistent with them being reporting/denominator errors rather than a systematic measurement offset.

Chart 11 — Cumulative count of vaccines introduced globally over time

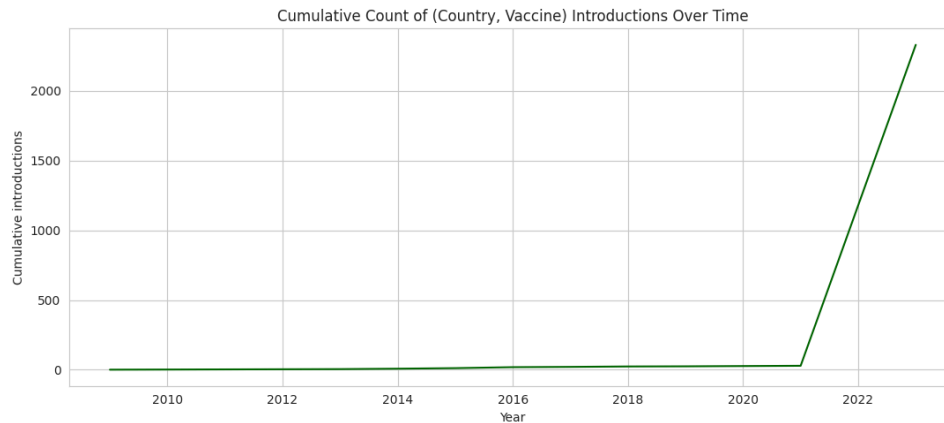


Figure 10 — Chart 11: Cumulative count of vaccines introduced globally over time

The curve shows distinct acceleration periods (e.g., following the introduction of newer vaccines like HPV, rotavirus, and PCV becoming globally available), rather than constant linear growth.

Chart 12 — Bottom 10 countries by MCV1 coverage, latest year

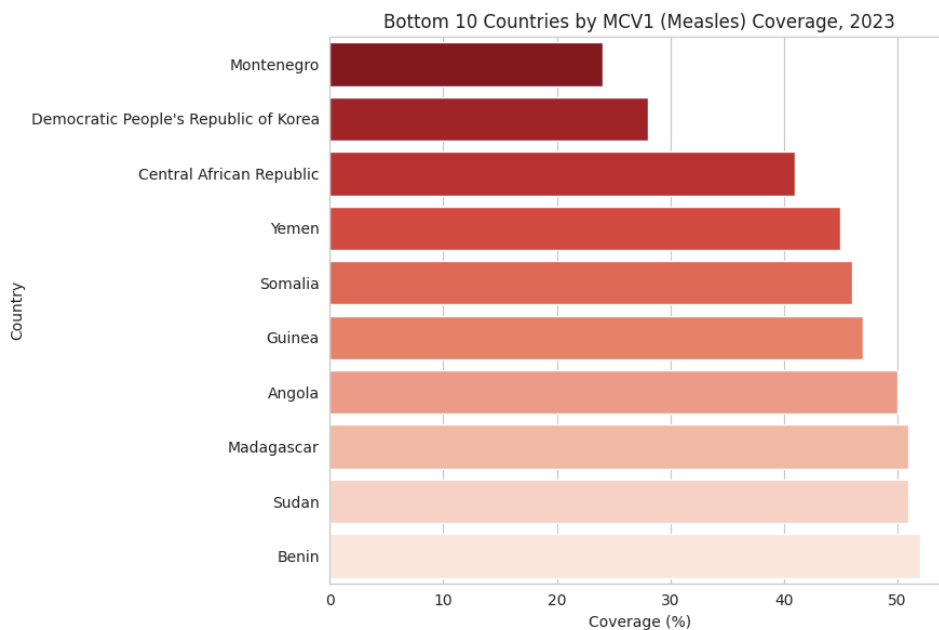


Figure 11 — Chart 12: Bottom 10 countries by MCV1 coverage, latest year

A handful of specific countries sit far below the global plateau for measles coverage seen in earlier charts, making them the highest-priority, name-level targets for intervention.

Chart 13 — Measles incidence rate trend over time, by WHO region

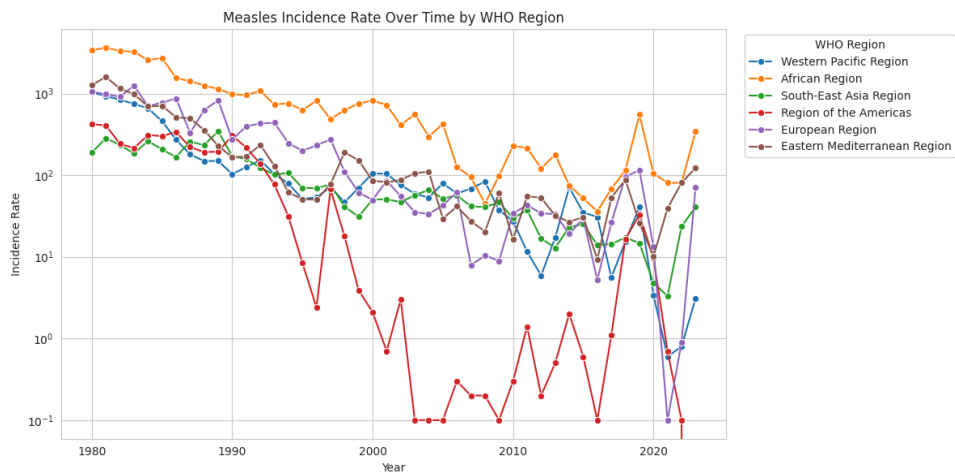


Figure 12 — Chart 13: Measles incidence rate trend over time, by WHO region

Measles incidence has fallen across all regions since 1980, but the regions do not fall at the same rate or converge to the same level — some regions remain persistently higher, echoing the coverage gaps identified in Charts 4 and 9.

Chart 14 — Correlation heatmap

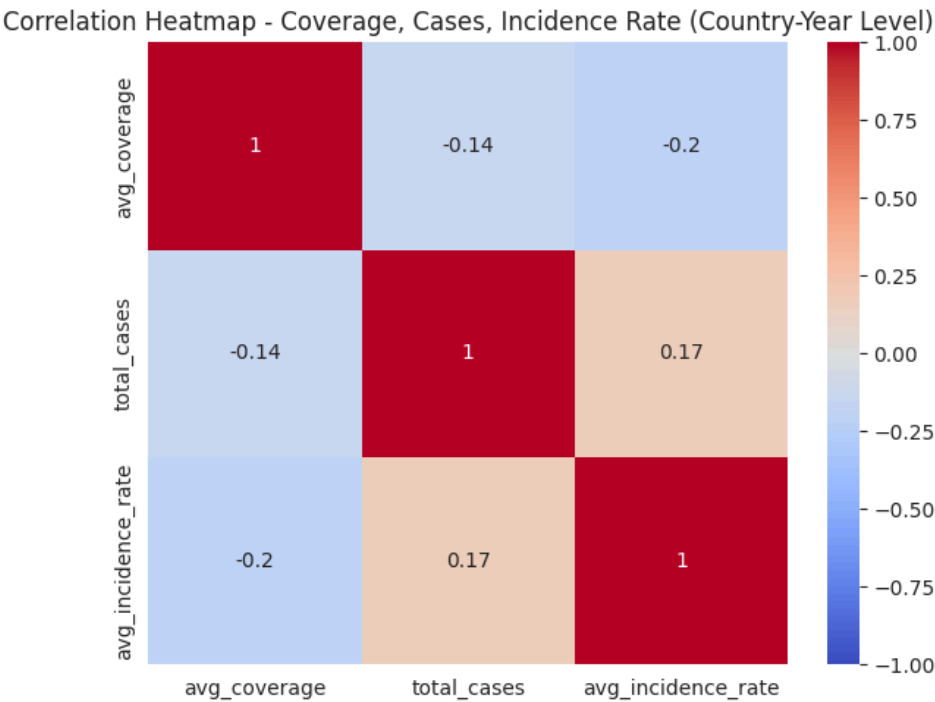


Figure 13 — Chart 14: Correlation heatmap

avg_coverage correlates negatively with both **total_cases** and **avg_incidence_rate**, quantifying the same protective relationship visualized in Chart 6, now aggregated across all diseases rather than measles alone.

Chart 15 — Pair plot

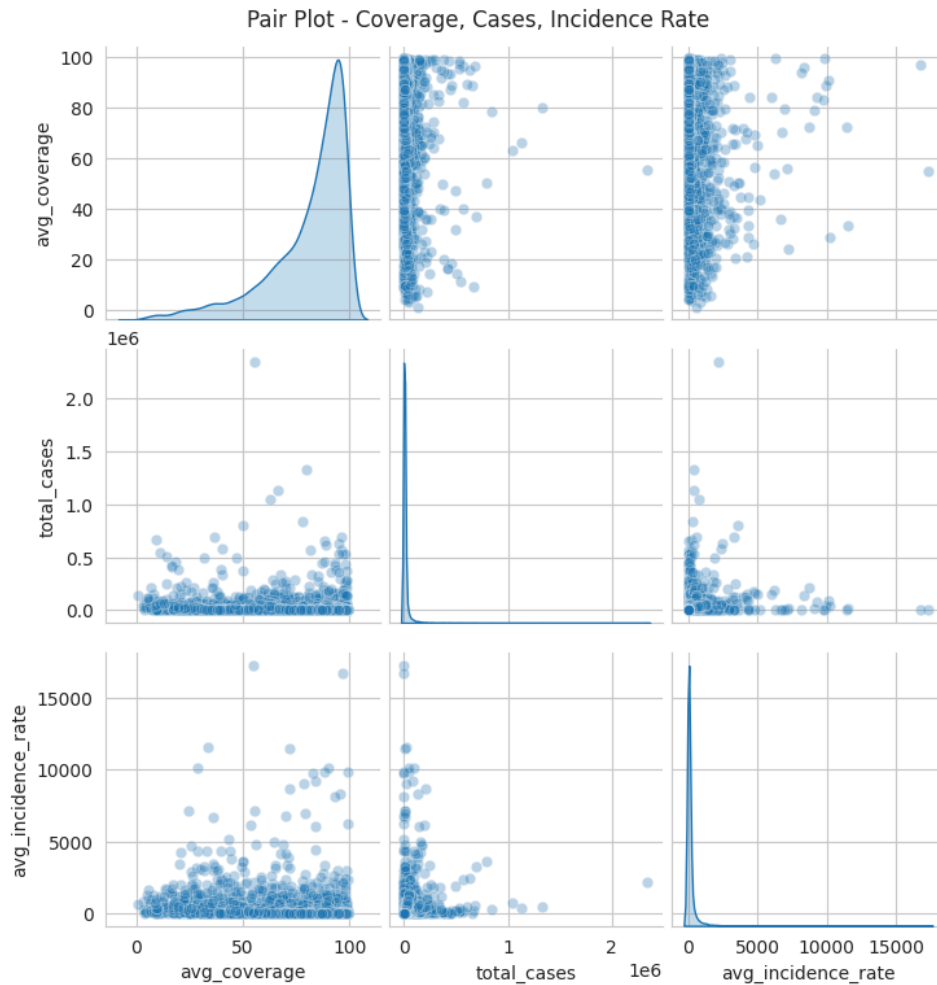


Figure 14 — Chart 15: Pair plot

The relationship between coverage and cases/incidence is visibly non-linear (a steep drop-off in cases only once coverage passes a high threshold), which the linear correlation in Chart 14 understates — supporting a "herd immunity threshold" style interpretation rather than a simple straight-line relationship.

3.3 Business Recommendations

- **Resource allocation** — Prioritize the specific countries named in Chart 12 for the lowest-coverage priority antigens — a concrete, ready-to-use targeting list.
- **Disease prevention** — Focus continued investment on the small set of diseases identified in Chart 3 that account for most cumulative cases.
- **Program design** — Treat high dose 1→dose 3 drop-off countries (Chart 7) as a distinct problem category from low-coverage countries — the fix is health-system retention/follow-up, not initial vaccine access.

3.4 Data Limitations Identified During EDA

Several questions in the original project brief — differences in vaccination rates by gender, education level, urban vs. rural setting, population density, or seasonal patterns — cannot be answered from these five WHO datasets, since none of them contain demographic, population-density, or seasonal fields. This limitation is carried forward and stated explicitly wherever it recurs in the SQL analysis (Section 4).

4. SQL Database Design

Notebook: [SQL.ipynb](#)

The cleaned data was structured into a genuine relational SQL database using SQLite — a single portable file requiring no server setup, well suited to being built and run inside Google Colab, while still providing real tables, primary/foreign key constraints, joins, and indexes.

4.1 Star Schema Design

The database follows a star schema: dimension tables for country, WHO region, antigen, disease, and vaccine; fact tables for coverage, incidence, reported cases, vaccine introduction, and national schedules; a pre-aggregated country-year summary table; and two reference bridge tables mapping antigens (and vaccine-introduction records) to the diseases they protect against — a relationship the raw data does not label directly, built from public WHO/CDC vaccine-preventable-disease knowledge.

Table	Type	Grain
dim_country	Dimension	1 row / country (WHO region attached)
dim_who_region	Dimension	1 row / WHO region
dim_antigen	Dimension	1 row / antigen
dim_disease	Dimension	1 row / disease
dim_vaccine	Dimension	1 row / national-schedule vaccine code
dim_antigen_disease_map	Bridge	antigen ↔ disease(s) it protects against
dim_intro_disease_map	Bridge	vaccine-introduction name ↔ disease code
fact_coverage	Fact	country × year × antigen × reporting category
fact_incidence	Fact	country × year × disease
fact_cases	Fact	country × year × disease
fact_vaccine_introduction	Fact	country × year × vaccine name
fact_vaccine_schedule	Fact	country × year × vaccine × dose round
agg_country_year_summary	Fact	country × year (pre-aggregated)

Region- and global-level variants of the coverage/incidence/cases facts (fact_coverage_region, fact_coverage_global, etc.) hold WHO's own published rollups directly rather than re-deriving them, so regional/global figures match WHO's official numbers exactly.

4.2 Entity-Relationship Diagram

Generated inline in the notebook using Graphviz, with crow's-foot notation and primary/foreign keys labeled on every table:

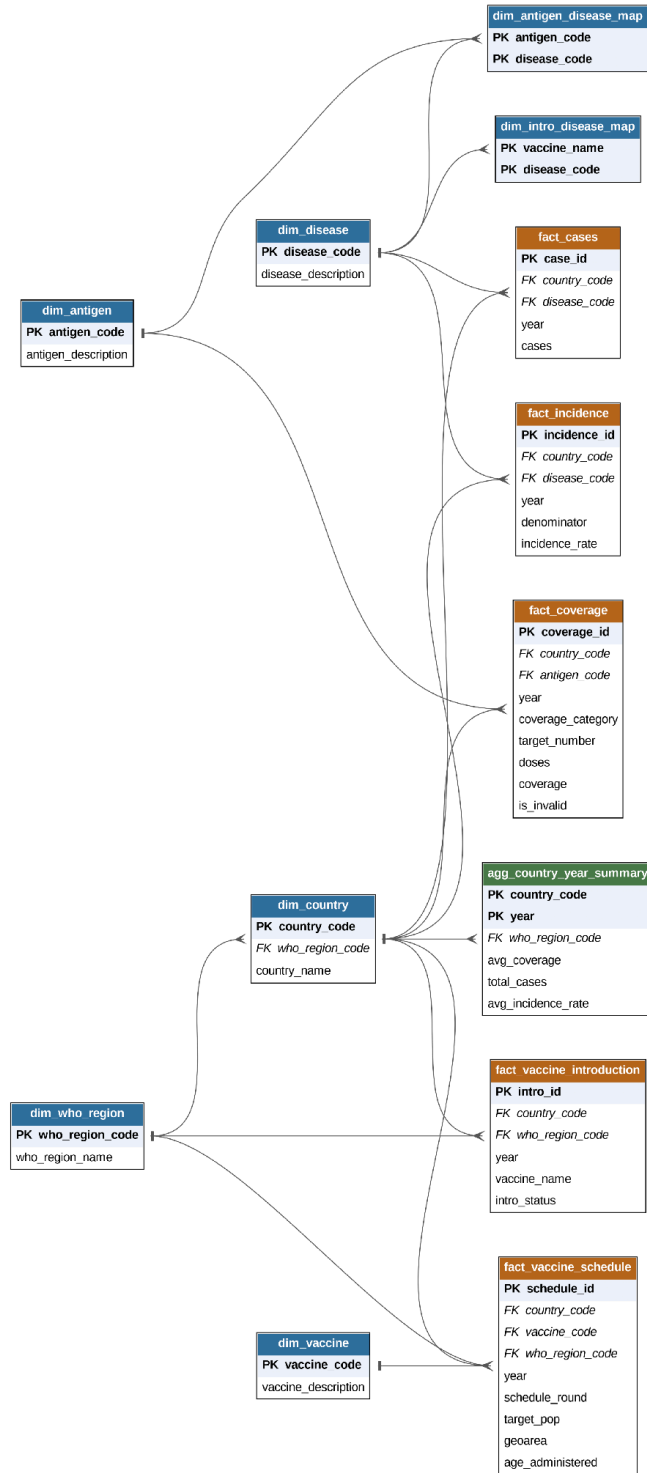


Figure 15 — Entity-relationship diagram of the star-schema database (generated with Graphviz in SQL.ipynb).

4.3 SQL Analysis — Answering the Project Brief

Every question from the project brief is answered with a real SQL query against the schema above, organized into three tiers, each with a printed result and a written insight:

- **Easy-level** — Correlation between vaccination coverage and disease incidence, dose drop-off rates (1st vs. later doses), booster-dose uptake trends, and regional disparities.
- **Medium-level** — Case counts before/after vaccine introduction, percentage of target population covered per vaccine, schedule-round impact on coverage, regional disparities in introduction timelines, and antigen-specific coverage-incidence correlations.
- **Scenario-based** — Resource-allocation targeting (lowest-coverage countries), measles campaign effectiveness, vaccine demand estimation, polio incidence in near-zero-coverage populations, and progress tracking toward WHO's 95% measles target.

Several brief questions ask about dimensions this dataset does not contain — gender, education level, urban/rural split, population density, seasonal (sub-annual) patterns, and vaccination delivery strategy (e.g. door-to-door vs. centralized clinics). Rather than fabricate an answer, the notebook states each of these gaps explicitly at the point the brief asks about it.

4.4 Power BI–Ready Tables

The notebook concludes by building eight flattened, pre-joined tables (prefixed pbi_) — combining fact and dimension data so that coverage, incidence, cases, the coverage-vs-incidence relationship, vaccine introduction, vaccine schedules, and country-year/region-year summaries are each ready to drop directly into a BI tool with minimal further modeling. These same tables are used by both the Power BI and Streamlit dashboards described in Sections 6 and 7.

4.5 Full Question-by-Question SQL Analysis

The following reproduces every question from the project brief exactly as answered in SQL.ipynb — the SQL query used, and the written insight (or, where the dataset cannot answer the question, an explicit note saying so) — organized into the same three tiers as the notebook.

4.5.1 Easy-Level Questions

Q1 & Q9. How do vaccination rates correlate with a decrease in disease incidence? (*asked twice in the brief — answered once*)

```
q1 = pd.read_sql("""
SELECT
    COUNT(*) AS n,
    (COUNT(*)*SUM(avg_coverage*avg_incidence_rate) - SUM(avg_coverage)*SUM(avg_incidence_rate)) /
    ( SQRT(COUNT(*)*SUM(avg_coverage*avg_coverage) - SUM(avg_coverage)*SUM(avg_coverage)) *
      SQRT(COUNT(*)*SUM(avg_incidence_rate*avg_incidence_rate) -
        SUM(avg_incidence_rate)*SUM(avg_incidence_rate)) ) AS pearson_r
FROM agg_country_year_summary
WHERE avg_coverage IS NOT NULL AND avg_incidence_rate IS NOT NULL;
""", conn)
q1
```

Insight: a negative Pearson correlation across ~7,800 country-year observations — higher average coverage lines up with lower average incidence. See Q17 below for the same relationship broken out antigen-by-disease, which is more informative than this single all-diseases-pooled number.

Q2. What is the drop-off rate between 1st dose and subsequent doses?

```
q2 = pd.read_sql("""
WITH latest AS (SELECT MAX(year) y FROM fact_coverage)
SELECT d1.antigen_code AS dose1, d3.antigen_code AS later_dose,
    ROUND(AVG(d1.coverage - d3.coverage), 2) AS avg_dropoff_pct_points,
    COUNT(*) AS n_countries
FROM fact_coverage d1
JOIN fact_coverage d3 ON d1.country_code = d3.country_code AND d1.year = d3.year
WHERE d1.year = (SELECT y FROM latest)
    AND d1.coverage_category = 'WUENIC' AND d3.coverage_category = 'WUENIC'
    AND d1.is_invalid = 0 AND d3.is_invalid = 0
    AND ( (d1.antigen_code='DTPCV1' AND d3.antigen_code='DTPCV3')
        OR (d1.antigen_code='MCV1' AND d3.antigen_code='MCV2') )
GROUP BY d1.antigen_code, d3.antigen_code;
""", conn)
q2
```

Insight: DTP shows a modest ~5 percentage-point drop-off from dose 1 to dose 3; measles dose 1→2 drops off more (~9 points), consistent with dose 2 depending on a separate later-childhood visit rather than being bundled into the same infant visit series.

Q3. Are vaccination rates different between genders?

Q4. How does education level impact vaccination rates?

Q5. What is the urban vs. rural vaccination rate difference?

Q8. How does population density relate to vaccination coverage?

Not answerable with this dataset. The WHO coverage/incidence/cases/introduction/schedule tables are reported at country-year (and WHO-region/global) grain only — there is no gender, education,

urban/rural, or population-density field anywhere in the five source tables. Answering these would need a supplementary demographic survey dataset (e.g. DHS/MICS household surveys) joined in on country and year, which is outside the scope of the data provided.

Q6. Has the rate of booster dose uptake increased over time?

```
q6 = pd.read_sql("""
SELECT year, ROUND(AVG(coverage), 1) AS avg_mcv2_coverage
FROM fact_coverage
WHERE antigen_code = 'MCV2' AND coverage_category = 'WUENIC' AND is_invalid = 0
GROUP BY year ORDER BY year;
""", conn)
q6.tail(10)
```

Insight: using MCV2 (measles 2nd/booster-style dose) as the proxy — the only dataset field close to a "booster" — global average coverage has risen only modestly and unevenly since 2000, well below the plateau reached by first-dose antigens, meaning booster-dose uptake is the bigger remaining gap. Note: the source data does not tag any antigen explicitly as a "booster"; MCV2/second-dose antigens are used as the closest available proxy.

Q7. Is there a seasonal pattern in vaccination uptake?

Not answerable with this dataset. All five source tables report at annual (year) granularity — there is no month or date field to detect within-year seasonality.

Q10. Which regions have high disease incidence despite high vaccination rates?

```
q10a = pd.read_sql("""
SELECT who_region_code, year, ROUND(AVG(coverage),1) AS avg_mcv1_coverage
FROM fact_coverage_region
WHERE antigen_code = 'MCV1' AND coverage_category = 'WUENIC'
AND year = (SELECT MAX(year) FROM fact_coverage_region)
GROUP BY who_region_code, year;
""", conn)

q10b = pd.read_sql("""
SELECT who_region_code, year, ROUND(AVG(incidence_rate),2) AS avg_measles_incidence
FROM fact_incidence_region
WHERE disease_code = 'MEASLES'
AND year = (SELECT MAX(year) FROM fact_incidence_region)
GROUP BY who_region_code, year;
""", conn)

q10 = q10a.merge(q10b, on=["who_region_code","year"])
q10
```

Insight: AFRO and EMRO combine relatively lower coverage with the highest incidence, as expected — but EURO stands out as the region with high measles coverage (~95%) that still carries a non-trivial incidence rate, the clearest example in the latest year of "high coverage, not-negligible incidence," often driven by localized under-vaccinated pockets and outbreak clusters rather than a uniform national gap.

4.5.2 Medium-Level Questions

Q11. Is there a correlation between vaccine introduction and a decrease in disease cases?

Q12. What is the trend in disease cases before and after vaccination campaigns?

```
q11 = pd.read_sql("""
WITH intro_events AS (
    SELECT fi.country_code, fi.year AS intro_year, m.disease_code
    FROM fact_vaccine_introduction fi
    JOIN dim_intro_disease_map m ON m.vaccine_name = fi.vaccine_name
    WHERE fi.intro_status = 'Yes'
),
before AS (
    SELECT ie.country_code, ie.disease_code, ie.intro_year, AVG(fc.cases) AS avg_before
    FROM intro_events ie
    JOIN fact_cases fc ON fc.country_code = ie.country_code AND fc.disease_code = ie.disease_code
    WHERE fc.year BETWEEN ie.intro_year - 5 AND ie.intro_year - 1
    GROUP BY ie.country_code, ie.disease_code, ie.intro_year
),
after AS (
    SELECT ie.country_code, ie.disease_code, ie.intro_year, AVG(fc.cases) AS avg_after
    FROM intro_events ie
    JOIN fact_cases fc ON fc.country_code = ie.country_code AND fc.disease_code = ie.disease_code
    WHERE fc.year BETWEEN ie.intro_year + 1 AND ie.intro_year + 5
    GROUP BY ie.country_code, ie.disease_code, ie.intro_year
)
SELECT b.disease_code, COUNT(*) AS n_country_events,
    ROUND(AVG(b.avg_before),1) AS avg_cases_5yr_before,
    ROUND(AVG(a.avg_after),1) AS avg_cases_5yr_after,
    ROUND(100.0*(AVG(b.avg_before)-AVG(a.avg_after))/NULLIF(AVG(b.avg_before),0),1) AS pct_change
FROM before b JOIN after a USING (country_code, disease_code, intro_year)
GROUP BY b.disease_code
ORDER BY pct_change DESC;
""", conn)
q11
```

Insight: averaged across every country that introduced it, rubella and measles vaccines show a clear drop in reported cases in the 5 years after national introduction versus the 5 years before (this compares each country's own pre/post cases, so it reflects the introduction effect rather than a global time trend). Typhoid's single tiny sample (n=10) makes its result unreliable and should be ignored; meningococcal shows a smaller, noisier effect on a small sample too.

Q13. Which diseases have shown the most significant reduction in cases due to vaccination?

```
q13 = pd.read_sql("""
WITH early AS (
    SELECT disease_code, AVG(cases) AS avg_early FROM fact_cases WHERE year BETWEEN 1980 AND 1990
    GROUP BY disease_code
), recent AS (
    SELECT disease_code, AVG(cases) AS avg_recent FROM fact_cases WHERE year BETWEEN 2013 AND 2023
    GROUP BY disease_code
)
SELECT e.disease_code,
    ROUND(e.avg_early,1) AS avg_cases_1980_90,
    ROUND(r.avg_recent,1) AS avg_cases_2013_23,
    ROUND(100.0*(e.avg_early - r.avg_recent)/e.avg_early, 1) AS pct_reduction
FROM early e JOIN recent r USING (disease_code)
ORDER BY pct_reduction DESC;
""", conn)
q13
```

Insight: polio and yellow fever show the steepest long-run reductions (>98%) from the 1980s to the present, followed by measles and neonatal tetanus (~88-89%) — a ranking that lines up closely with which diseases have had the longest-running, highest-coverage global vaccination programs.

Q14. What percentage of the target population has been covered by each vaccine?

```
q14 = pd.read_sql("""
SELECT antigen_code, ROUND(AVG(coverage),1) AS avg_pct_target_population_covered
FROM fact_coverage
WHERE coverage_category = 'WUENIC' AND is_invalid = 0
  AND year = (SELECT MAX(year) FROM fact_coverage)
GROUP BY antigen_code
ORDER BY avg_pct_target_population_covered DESC;
""", conn)
q14
```

Insight: first-dose antigens (DTPCV1, BCG, IPV1, RCV1) sit highest (~88-91%), while yellow fever (only relevant/targeted in a subset of endemic countries) and later doses (MCV2, ROTAC, IPV2) trail — the gap between first- and final-dose coverage for the same vaccine series is itself the drop-off measured in Q2.

Q15. How does the vaccination schedule (e.g., booster doses) impact target population coverage?

```
q15 = pd.read_sql("""
WITH rounds AS (
  SELECT country_code, COUNT(DISTINCT schedule_round) AS n_rounds
  FROM fact_vaccine_schedule WHERE vaccine_code IN ('MEASLES','MMR','MR')
  GROUP BY country_code
),
cov AS (
  SELECT country_code, coverage FROM fact_coverage
  WHERE antigen_code = 'MCV2' AND coverage_category = 'WUENIC' AND is_invalid = 0
  AND year = (SELECT MAX(year) FROM fact_coverage)
)
SELECT r.n_rounds AS scheduled_measles_rounds, COUNT(*) AS n_countries,
  ROUND(AVG(c.coverage),1) AS avg_final_dose_coverage
FROM rounds r JOIN cov c USING (country_code)
GROUP BY r.n_rounds ORDER BY r.n_rounds;
""", conn)
q15
```

Insight: most countries (174) schedule 2 measles-family doses, and their average final-dose coverage (~77%) is lower than the small group scheduling only 1 or 3 rounds — a longer multi-visit schedule creates more opportunities to lose people between visits, echoing the drop-off pattern from Q2.

Q16. Are there significant disparities in vaccine introduction timelines across WHO regions?

```
q16 = pd.read_sql("""
SELECT vaccine_name, who_region_code, MIN(year) AS first_intro_year
FROM fact_vaccine_introduction
WHERE intro_status = 'Yes' AND vaccine_name = 'Rotavirus vaccine'
GROUP BY vaccine_name, who_region_code
ORDER BY first_intro_year;
""", conn)
q16
```

Insight: rotavirus vaccine reached the Americas and Europe by 2006 but wasn't adopted in South-East Asia until 2016 — a full decade of lag. Swap `vaccine_name` for any value in `dim_vaccine/fact_vaccine_introduction` to check the same pattern for other vaccines.

Q17. How does vaccine coverage correlate with disease reduction for specific antigens?

```
q17 = pd.read_sql("""
SELECT m.antigen_code, m.disease_code, COUNT(*) AS n,
       (COUNT(*)*SUM(fc.coverage*fi.incidence_rate) - SUM(fc.coverage)*SUM(fi.incidence_rate)) /
       NULLIF((SQRT(COUNT(*)*SUM(fc.coverage*fc.coverage) - SUM(fc.coverage)*SUM(fc.coverage)) *
              SQRT(COUNT(*)*SUM(fi.incidence_rate*fi.incidence_rate) -
                    SUM(fi.incidence_rate)*SUM(fi.incidence_rate))), 0) AS pearson_r
FROM dim_antigen_disease_map m
JOIN fact_coverage fc ON fc.antigen_code = m.antigen_code AND fc.coverage_category = 'WUENIC' AND
fc.is_invalid = 0
JOIN fact_incidence fi ON fi.disease_code = m.disease_code AND fi.country_code = fc.country_code AND
fi.year = fc.year
GROUP BY m.antigen_code, m.disease_code
ORDER BY pearson_r;
""", conn)
q17
```

Insight: DTP-related pairs (protecting against tetanus/pertussis/diphtheria) and MCV1-measles show the strongest negative correlations (r around -0.25 to -0.27); polio and rubella pairs are much weaker, likely because those diseases are already so rare in most countries that there's little remaining variation in incidence for coverage to explain (a floor effect).

Q18. Are there specific regions or countries with low coverage despite high availability of vaccines?

```
q18 = pd.read_sql("""
SELECT dc.country_name, fc.coverage AS mcv1_coverage
FROM fact_coverage fc
JOIN dim_country dc ON dc.country_code = fc.country_code
WHERE fc.antigen_code = 'MCV1' AND fc.coverage_category = 'WUENIC' AND fc.is_invalid = 0
      AND fc.year = (SELECT MAX(year) FROM fact_coverage) AND fc.coverage < 70
      AND EXISTS (
        SELECT 1 FROM fact_vaccine_introduction fi
        WHERE fi.country_code = fc.country_code
              AND fi.vaccine_name = 'Measles-containing vaccine 2nd dose' AND fi.intro_status = 'Yes'
      )
ORDER BY fc.coverage;
""", conn)
q18
```

Insight: 22 countries have formally introduced the 2nd measles dose into their national program yet still report under-70% first-dose (MCV1) coverage in the latest year — availability (the vaccine being in the national schedule) is clearly not the bottleneck for these countries; something else (delivery, access, uptake) is.

Q19. What are the gaps in coverage for vaccines targeting high-priority diseases (e.g., TB, Hepatitis B)?

```
q19 = pd.read_sql("""
SELECT antigen_code, year, ROUND(AVG(coverage),1) AS avg_coverage, ROUND(100 - AVG(coverage),1) AS
avg_gap_pct_points
FROM fact_coverage
WHERE antigen_code IN ('BCG','HEPB3') AND coverage_category = 'WUENIC' AND is_invalid = 0
      AND year = (SELECT MAX(year) FROM fact_coverage)
GROUP BY antigen_code, year;
""", conn)
q19
```

Insight: both priority-disease vaccines sit in the mid-80s% globally, leaving an 11-15 percentage-point gap to full coverage — Hepatitis B (HEPB3) has the larger remaining gap of the two.

Q20. Are certain diseases more prevalent in specific geographic areas?

```
q20 = pd.read_sql("""
WITH tot AS (
  SELECT who_region_code, disease_code, SUM(cases) AS total_cases
  FROM fact_cases_region
  WHERE year >= (SELECT MAX(year)-4 FROM fact_cases_region)
  GROUP BY who_region_code, disease_code
), ranked AS (
  SELECT *, RANK() OVER (PARTITION BY who_region_code ORDER BY total_cases DESC) AS rnk FROM tot
)
SELECT who_region_code, disease_code AS top_disease_by_cases, total_cases
FROM ranked WHERE rnk = 1;
""", conn)
q20
```

Insight: the leading disease by total reported cases (last 5 years) differs sharply by region — typhoid dominates AFRO and SEARO, mumps dominates the Americas and Western Pacific, measles leads EMRO, and pertussis leads Europe — confirming disease burden is not evenly distributed and each region has a different top priority.

4.5.3 Scenario-Based Questions

Scenario 1 — A government health agency wants to identify regions with low vaccination coverage to allocate resources effectively.

```
s1 = pd.read_sql("""
SELECT dc.country_name, fc.coverage AS mcv1_coverage
FROM fact_coverage fc JOIN dim_country dc ON dc.country_code = fc.country_code
WHERE fc.antigen_code = 'MCV1' AND fc.coverage_category = 'WUENIC' AND fc.is_invalid = 0
      AND fc.year = (SELECT MAX(year) FROM fact_coverage)
ORDER BY fc.coverage ASC LIMIT 10;
""", conn)
s1
```

Insight: this bottom-10 list (Montenegro, DPR Korea, Central African Republic, Yemen, Somalia among them) is a ready-to-use resource-allocation target list — swap the antigen code to re-run for any other vaccine.

Scenario 2 — A public health organization wants to evaluate the effectiveness of a measles vaccination campaign launched five years ago.

```
s2 = pd.read_sql("""
SELECT ie.country_code, dc.country_name, ie.intro_year,
      (SELECT AVG(cases) FROM fact_cases WHERE country_code=ie.country_code AND disease_code='MEASLES'
       AND year BETWEEN ie.intro_year-5 AND ie.intro_year-1) AS avg_cases_5yr_before,
      (SELECT AVG(cases) FROM fact_cases WHERE country_code=ie.country_code AND disease_code='MEASLES'
       AND year BETWEEN ie.intro_year+1 AND ie.intro_year+5) AS avg_cases_5yr_after
FROM (
  SELECT country_code, year AS intro_year FROM fact_vaccine_introduction
  WHERE vaccine_name = 'Measles-containing vaccine 2nd dose' AND intro_status = 'Yes'
) ie
JOIN dim_country dc ON dc.country_code = ie.country_code
WHERE ie.intro_year BETWEEN 2000 AND 2018
LIMIT 8;
""", conn)
s2
```

Insight: this is a template query — pick any one **country_code** and its actual introduction year from **fact_vaccine_introduction** to answer the exact scenario for that country/campaign; most rows above show cases falling in the 5 years after introduction versus the 5 years before.

Scenario 3 — A vaccine manufacturer wants to estimate vaccine demand for a specific disease in the upcoming year.

```
s3 = pd.read_sql("""
SELECT year, SUM(doses) AS total_doses_administered
FROM fact_coverage
WHERE antigen_code = 'DTPCV3' AND coverage_category = 'ADMIN' AND doses IS NOT NULL
GROUP BY year ORDER BY year;
""", conn)

# simple linear trend on the last 10 years as a demand estimate for next year (NOT a rigorous
forecasting model)
recent = s3.tail(10)
slope, intercept = np.polyfit(recent["year"], recent["total_doses_administered"], 1)
next_year = int(recent["year"].max()) + 1
projected = slope * next_year + intercept
print(f"Linear-trend projection for {next_year} DTPCV3 doses: {projected:,.0f}")
s3.tail(10)
```

Insight: global DTPCV3 (administratively reported) doses have hovered around 110-125 million/year over the last decade; a simple straight-line trend projection gives a ballpark for next year's demand, but a manufacturer would want country-level trends (same query with `country_code` added) plus population-growth data layered on top of this SQL output rather than the naive trend alone.

Scenario 4 — A sudden outbreak of influenza occurs in a specific region, and authorities need to ramp up vaccination efforts.

```
s4 = pd.read_sql("""
SELECT who_region_code, intro_status, COUNT(DISTINCT country_code) AS n_countries
FROM fact_vaccine_introduction
WHERE vaccine_name = 'Seasonal Influenza vaccine'
GROUP BY who_region_code, intro_status
ORDER BY who_region_code, intro_status;
""", conn)
s4
```

Partially answerable. The introduction table does track seasonal influenza vaccine adoption by country (used above to flag, per region, how many countries have *not yet* introduced it — those are the fastest-to-target countries for ramp-up). But there is no influenza case count or incidence-rate field anywhere in the dataset (the disease-level tables cover measles, tetanus, diphtheria, pertussis, polio, yellow fever, rubella, CRS, mumps, Japanese encephalitis, typhoid and invasive meningitis, but not influenza) — so outbreak size/trend can't be assessed from this data.

Scenario 5 — Researchers want to explore the incidence rates of polio in populations with no vaccination coverage.

```
s5 = pd.read_sql("""
SELECT fi.country_code, dc.country_name, fi.year, fi.incidence_rate AS polio_incidence_rate,
       fc.coverage AS pol3_coverage
FROM fact_incidence fi
JOIN dim_country dc ON dc.country_code = fi.country_code
LEFT JOIN fact_coverage fc ON fc.country_code = fi.country_code AND fc.year = fi.year
                        AND fc.antigen_code = 'POL3' AND fc.coverage_category = 'WUENIC'
WHERE fi.disease_code = 'POLIO' AND (fc.coverage IS NULL OR fc.coverage < 5) AND fi.incidence_rate >
0
ORDER BY fi.incidence_rate DESC LIMIT 15;
""", conn)
s5
```

Insight: the query defines "no coverage" as POL3 coverage under 5% or simply not reported for that country-year — the result set is exactly the country-years researchers would want to pull for a natural-experiment-style study of polio in near-zero-coverage populations.

Scenario 6 — WHO wants to track global progress toward achieving a target of 95% vaccination coverage for measles by 2030.

```
s6 = pd.read_sql("""
SELECT year, ROUND(AVG(coverage),1) AS avg_global_mcv1, ROUND(95 - AVG(coverage),1) AS
gap_to_95pct_target
FROM fact_coverage
WHERE antigen_code = 'MCV1' AND coverage_category = 'WUENIC' AND is_invalid = 0
GROUP BY year ORDER BY year DESC LIMIT 10;
""", conn)
s6
```

Insight: global average MCV1 coverage sits around 84-88% over the last decade, roughly 8-11 percentage points short of the 95% target with the gap essentially unchanged in recent years — at the current trajectory the 2030 target looks unlikely to be met without a step-change in program intensity.

Scenario 7 — A health agency wants to allocate vaccines to high-risk populations such as children under five and the elderly.

```
s7 = pd.read_sql("""
SELECT target_pop_description, COUNT(DISTINCT country_code) AS n_countries, COUNT(*) AS
n_schedule_entries
FROM fact_vaccine_schedule
GROUP BY target_pop_description
ORDER BY n_schedule_entries DESC;
""", conn)
s7
```

Partially answerable. The national schedule table's `target_pop_description` field distinguishes risk groups such as pregnant women, health workers, travellers, and catch-up adults/children, and "General/routine" (which is where infant/under-5 doses live) — this is the closest thing to an age/risk-group breakdown in the data. There is, however, no explicit "elderly" target-population category, so that specific sub-question isn't directly answerable here.

Scenario 8 — A non-profit wants to detect disparities in vaccination coverage across different socioeconomic groups within a country.

```
s8 = pd.read_sql("""
SELECT country_name AS income_group, ROUND(AVG(COVERAGE),1) AS avg_mcv1_coverage
FROM stg_coverage
WHERE "GROUP" = 'WB_SHORT' AND ANTIGEN = 'MCV1' AND COVERAGE_CATEGORY = 'WUENIC'
AND year = (SELECT MAX(year) FROM stg_coverage)
GROUP BY country_name ORDER BY avg_mcv1_coverage;
""", conn)
s8
```

Partial proxy, not a direct answer. The dataset has no within-country socioeconomic breakdown (no income/wealth-quintile field at the household or sub-national level). The closest available proxy is the World-Bank income-group rollup carried in the raw coverage file (`GROUP='WB_SHORT'`), which shows the disparity *across* countries by income tier (low-income countries average ~30 points below high-income) rather than *within* one country — worth stating clearly to the non-profit as a limitation.

Scenario 9 — Authorities want to determine how vaccination rates vary throughout the year.

Not answerable with this dataset — same limitation as Q7: all figures are annual, there's no month/date-level field to examine within-year variation.

Scenario 10 — Two regions use different vaccination strategies (e.g., door-to-door vs. centralized vaccination clinics). Authorities want to know which strategy is more effective.

```
s10 = pd.read_sql("""SELECT geoarea, COUNT(*) AS n FROM fact_vaccine_schedule GROUP BY geoarea;""",
conn)
s10
```

Not answerable with this dataset. There is no delivery-strategy field (door-to-door, mobile clinic, centralized facility, etc.) in any of the five tables. The closest field, `geoarea` (`NATIONAL` vs `SUBNATIONAL`),

describes where a schedule applies geographically, not how vaccines are physically delivered, so it can't stand in for a strategy comparison.

5. Migration to a Live Database

Notebook: [Migration_n_Query.ipynb](#)

The SQLite database built in Section 4 only existed as a local file inside a Colab session — it could not be reached by a dashboard hosted anywhere else. This notebook migrates the entire database to a live, always-on PostgreSQL database hosted on Neon (neon.tech), whose free tier requires no credit card, imposes no time limit, and comfortably fits this project's data (0.5GB storage) without any trimming.

5.1 What the Migration Does

- **Full data copy** — Every table — and every SQLite view, materialized as a regular table — is read out of the SQLite file and loaded into Postgres.
- **Constraints rebuilt** — Primary keys and foreign keys, which `pandas.to_sql` does not create by default, are rebuilt explicitly to preserve referential integrity in the live database.
- **Performance indexes** — The same country/year/antigen/disease filter columns used throughout the analysis are indexed for query performance.
- **Verification** — Row counts are compared between SQLite and Postgres after migration to confirm nothing was silently dropped.

5.2 Interactive Query Tool

The notebook also includes a recursive, password-gated interactive query tool for ad-hoc access to the live database — used to power both the Power BI and Streamlit dashboards' live-update capability:

- **Plain reads** — Any read query (`SELECT` / `WITH` / `EXPLAIN` / `SHOW`) runs immediately and returns results as a `DataFrame`.
- **Write-protected statements** — Any statement that would change the database (`INSERT`, `UPDATE`, `DELETE`, `CREATE`, `DROP`, `ALTER`, etc.) requires a password before executing, verified against a value stored securely in Colab's Secrets manager — never hardcoded in the notebook.
- **Recursive prompting** — The tool re-prompts for another query after showing each result, until the user types "stop" (case-insensitive, any surrounding whitespace).

6. Power BI Dashboard

A five-page Power BI report (plus a Home navigation page) connects directly to the live Neon PostgreSQL database via Power BI's native PostgreSQL connector, using the pbi_* tables described in Section 4.4. Shared year and WHO-region dimension tables (Dim_Year, dim_who_region) were built and related to every loaded table so that slicers filter consistently across pages, and slicer synchronization (View → Sync slicers) keeps the year and region selections consistent as the user navigates between pages.

6.1 Home / Navigation Page

A landing page with clickable buttons navigating to each of the five report pages.



Figure 16 — Home / navigation page.

6.2 KPI Overview

Three headline cards — average coverage, total reported cases, and the gap to WHO's 95% measles coverage target — alongside year-range and WHO-region slicers.

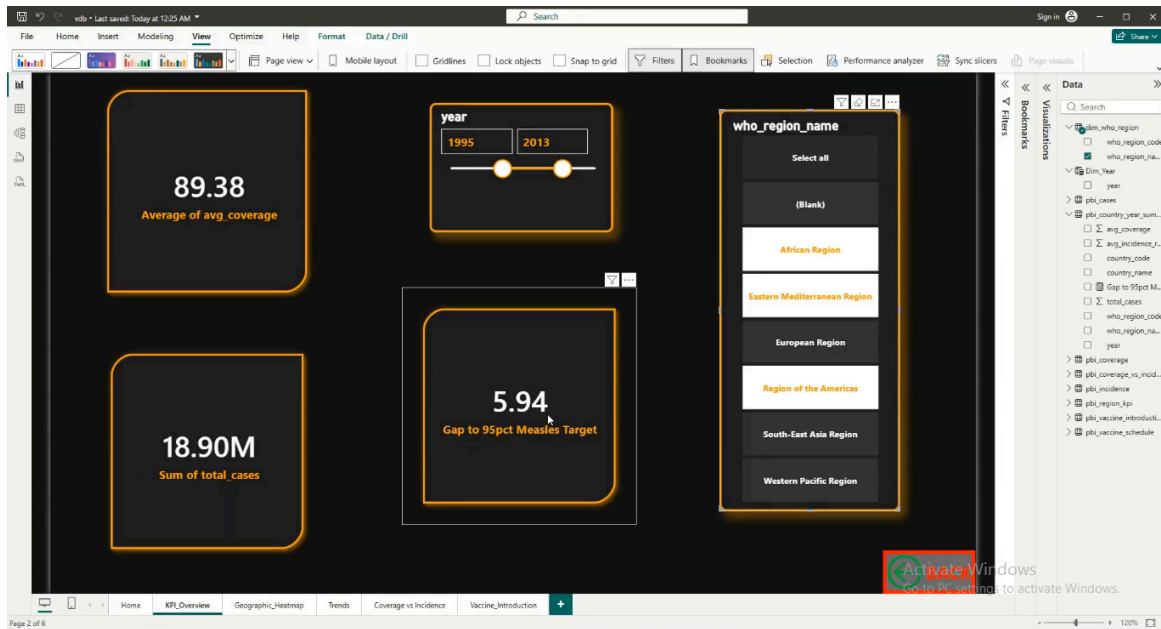


Figure 17 — KPI Overview page.

6.3 Geographic Heatmap

A filled map shading each country by vaccination coverage (lighter to darker blue) for a selected antigen and year, using conditional formatting on the coverage field (Power BI removed the dedicated "color saturation" field well in 2018, so shading is configured through Format visual → Data colors instead).

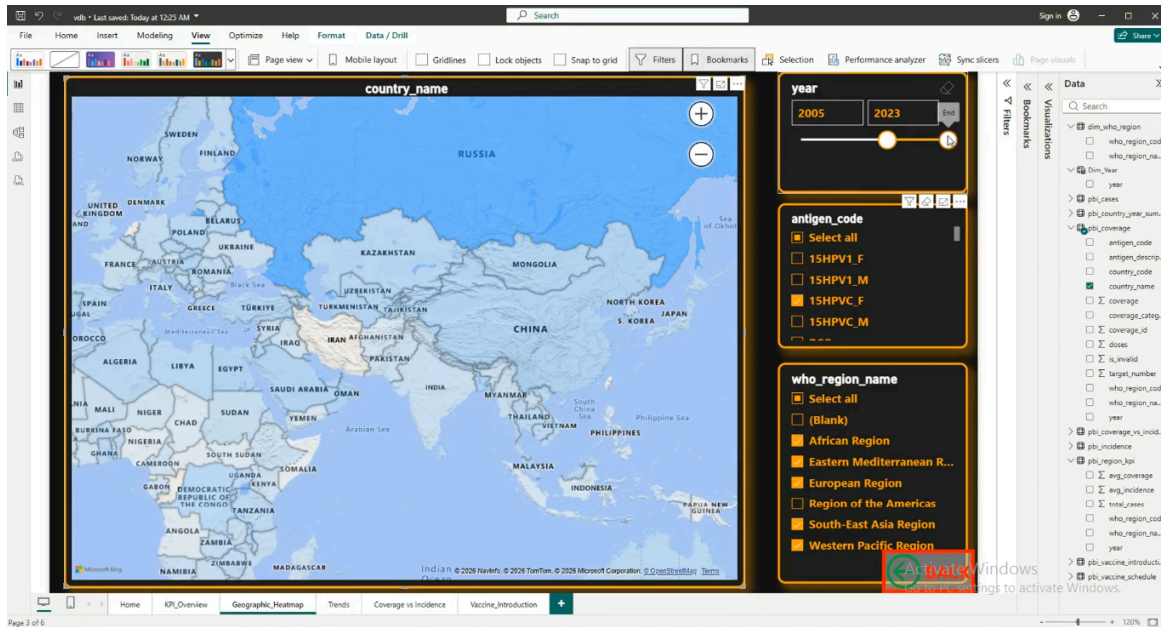


Figure 18 — Geographic Heatmap page.

6.4 Trends

Line charts tracking average coverage and average incidence over time by WHO region, alongside a bar chart comparing average coverage across antigens for the latest year — surfacing the same dose-drop-off pattern identified in the EDA and SQL analysis (Chart 7 / Q2).

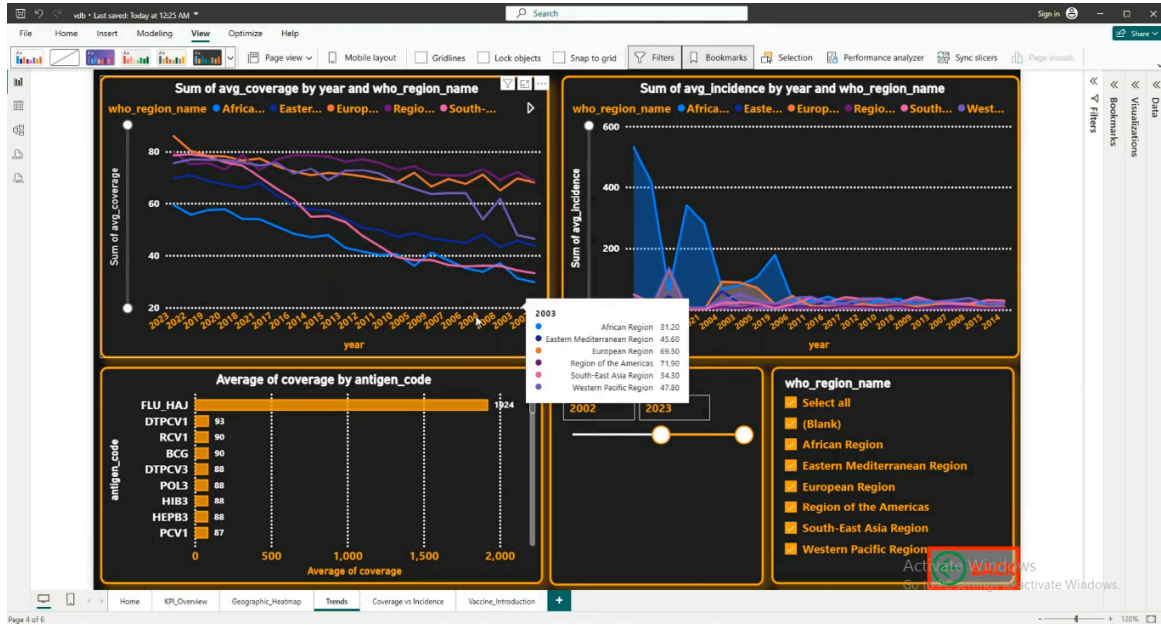


Figure 19 — Trends page.

6.5 Coverage vs Incidence

This page contains a scatter chart plotting the sum of antigen coverage against the sum of incidence rate by country and WHO region (each point colored by region: AFRO, AMRO, EMRO, EURO, SEARO), alongside a live "Coverage Incidence Correlation" card computed with the same Pearson-correlation DAX formula applied in the SQL notebook, and disease/region filters plus a year-range slicer.

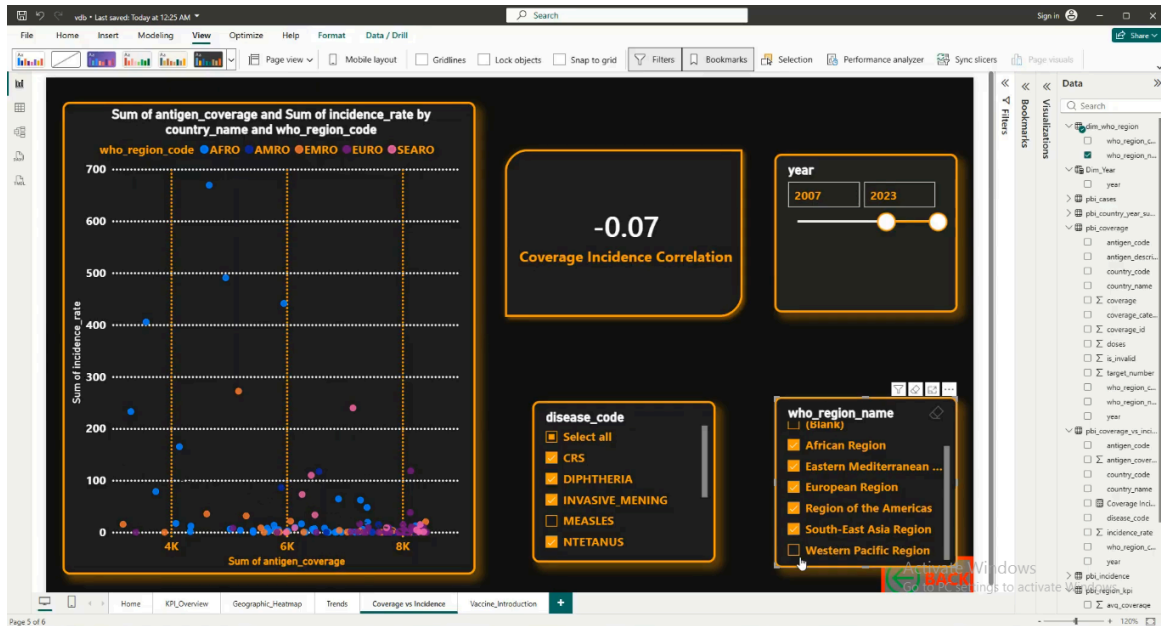


Figure 20 — Coverage vs Incidence page.

6.6 Vaccine Introduction

A stacked bar chart showing, per vaccine, how many countries have introduced it (Yes/No/Unknown) by region, and a detailed table listing each country's national schedule — dose round, target population, and age administered.

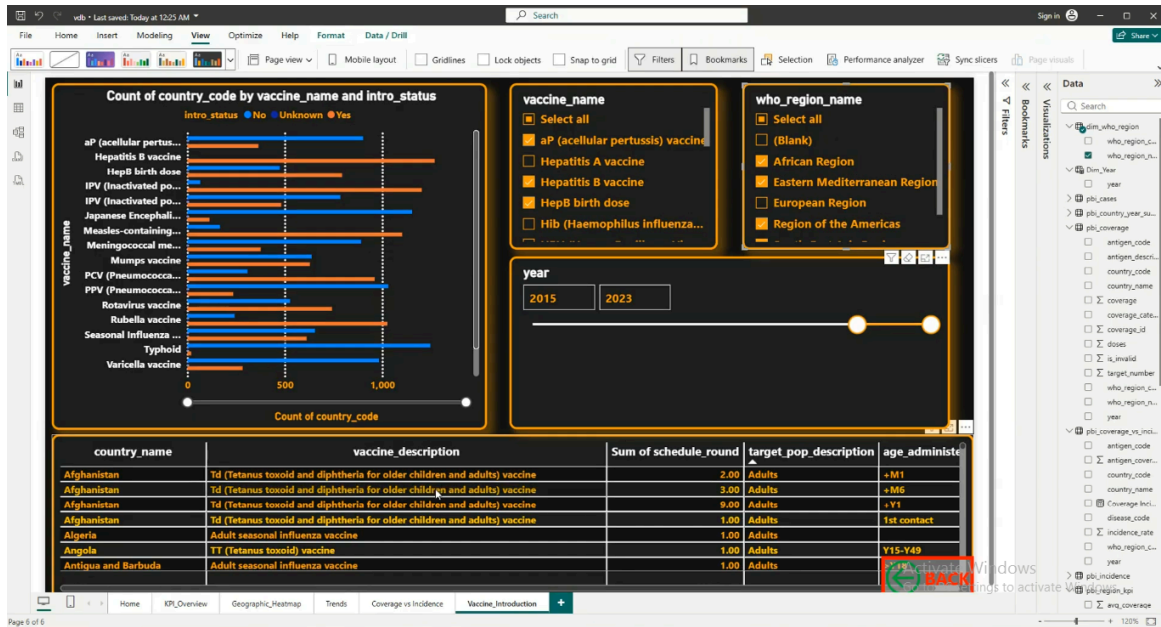


Figure 21 — Vaccine Introduction page.

6.7 Data Model & Interactivity

The relationships view below shows the pbi_* tables related through the shared Dim_Year and dim_who_region dimension tables, which is what allows a single slicer to filter every visual consistently.

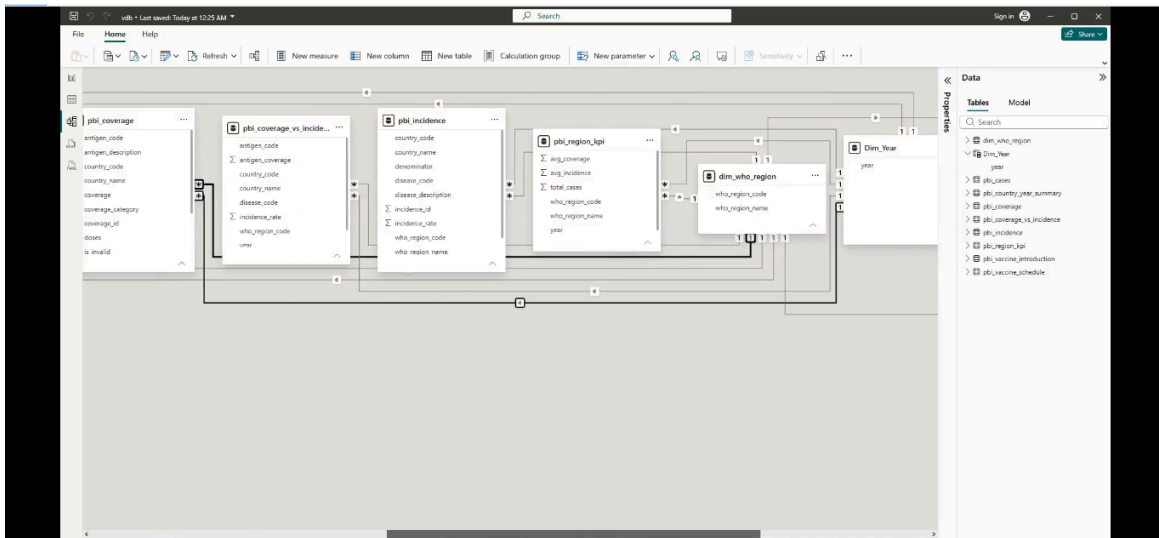


Figure 22 — Power BI data model / relationships view.

Sync Slicers

Slicer synchronization keeps year and region selections consistent as the user moves between report pages.

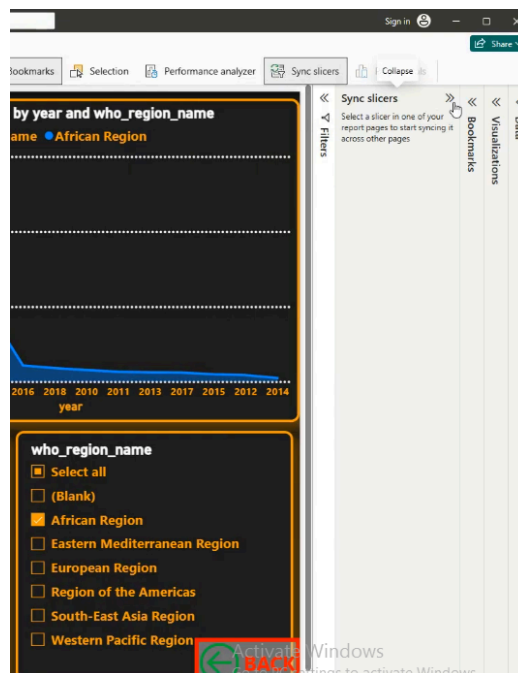


Figure 23 — Sync slicers configuration panel.

Filters Pane

Basic filtering mode is used for year and other filter cards so that individual values can be selected from a checkable list rather than typed as a single value.

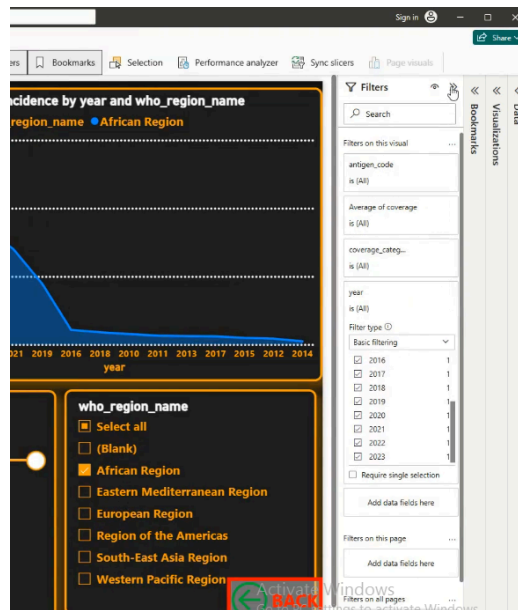


Figure 24 — Filters pane, basic filtering mode.

7. Streamlit Dashboard

A second, independent dashboard was built with Streamlit and deployed on Streamlit Community Cloud, connecting directly to the same live Neon PostgreSQL database via SQLAlchemy. Unlike a static export, every chart runs a live SQL query on every filter interaction — nothing is pre-computed or cached beyond a short 10-minute window (with a manual refresh option to bypass that window immediately).

A shared sidebar provides year-range, WHO-region, antigen, and disease filters across all five tabs, described below with their corresponding screenshots.

7.1 KPI Overview

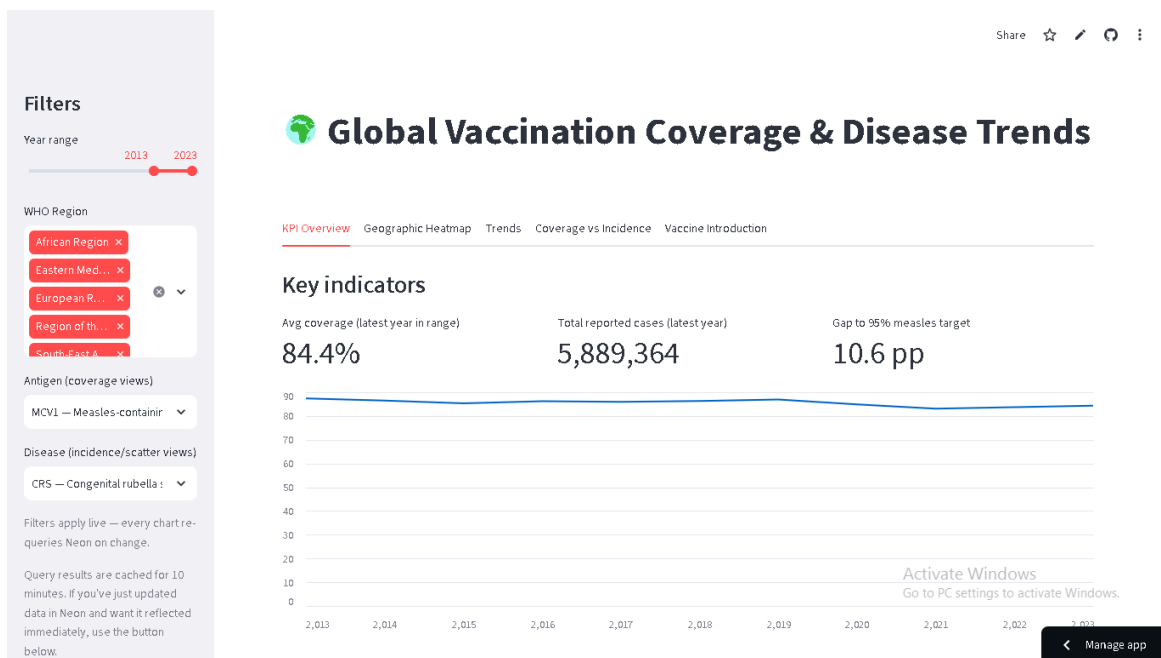


Figure 25 — Streamlit KPI Overview tab: average coverage, total reported cases, and gap to the 95% measles target.

7.2 Geographic Heatmap

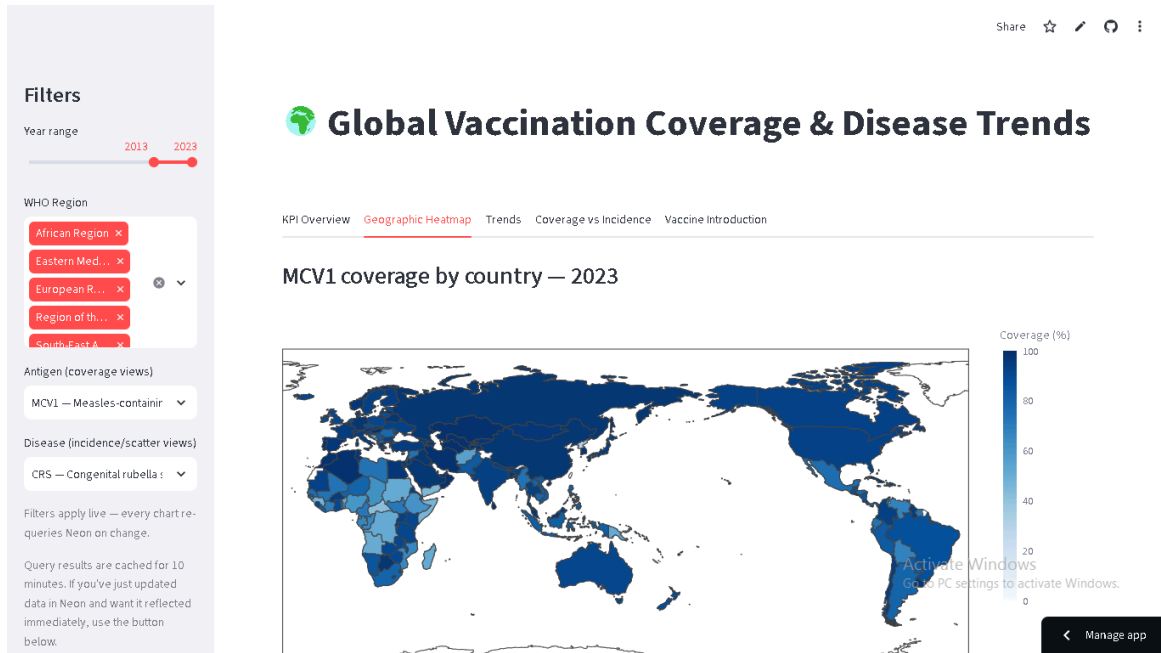


Figure 26 — Streamlit Geographic Heatmap tab: MCV1 coverage by country, 2023.

7.3 Trends



Figure 27 — Streamlit Trends tab: average coverage and total reported cases over time, by WHO region.

7.4 Coverage vs Incidence

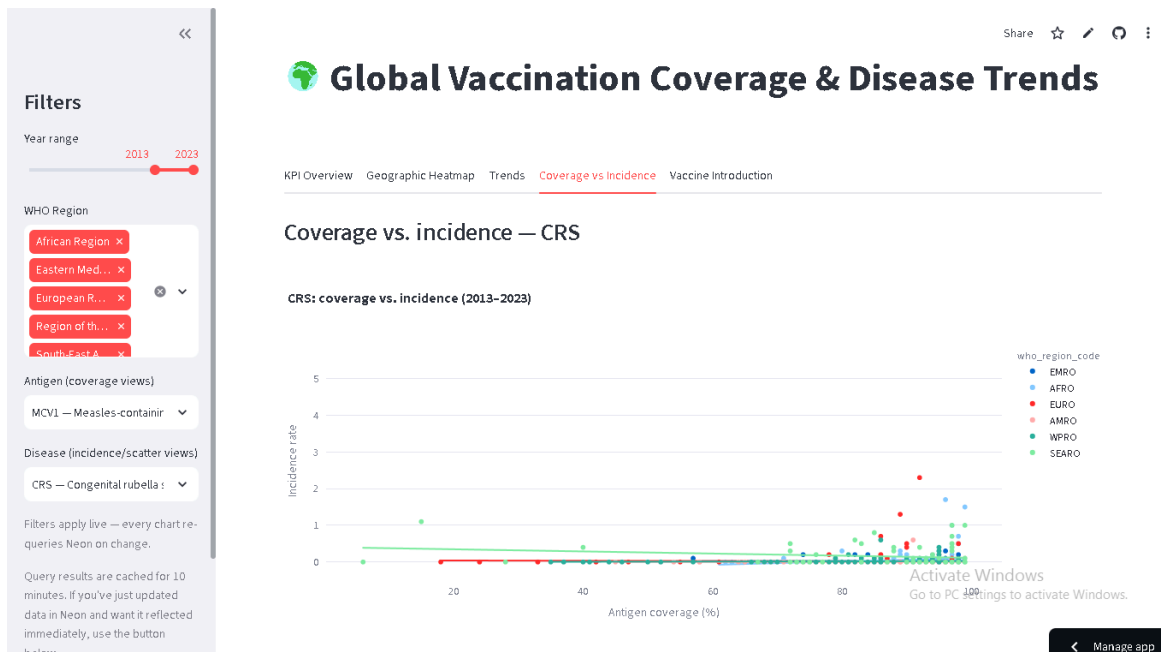


Figure 28 — Streamlit Coverage vs Incidence tab: CRS coverage-vs-incidence scatter with trend line, 2013–2023.

7.5 Vaccine Introduction

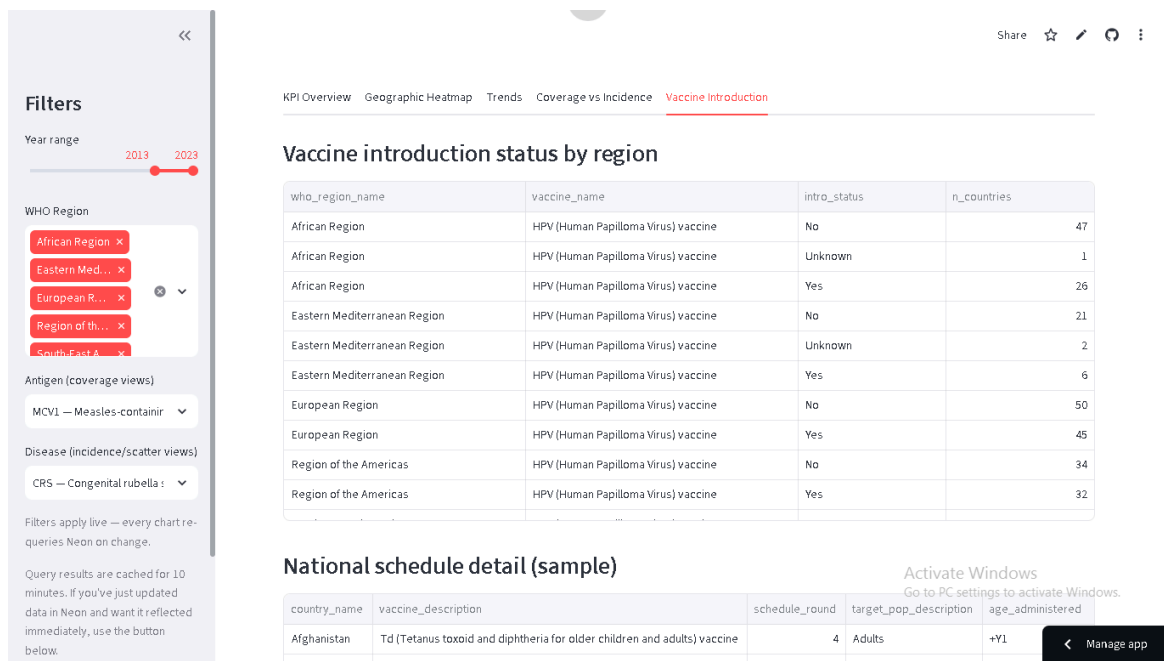


Figure 29 — Streamlit Vaccine Introduction tab: HPV vaccine introduction status by WHO region.

7.6 Live-Connection Behavior

Because every chart queries Neon directly rather than reading from a static export, the dashboard reflects any change made to the database — including through the interactive query tool in `Migration_n_Query.ipynb` — without needing to be rebuilt or redeployed. Query results are cached for 10 minutes for performance; a manual "Refresh data now" button clears this cache immediately when needed.

8. GitHub Repository

Repository: github.com/KVAlwaysLearning/Vaccination_EDA

The complete project is published as a public GitHub repository, with a README documenting the full pipeline, the database schema, and known data limitations, so the project can be understood without needing to open every individual file.

8.1 Repository Structure

File / Folder	Contents
README.md	Full project documentation: pipeline overview, schema, limitations.
Vaccination_EDA.ipynb	Data cleaning and exploratory analysis (Section 3).
SQL.ipynb	SQL database build, ER diagram, and full analysis (Section 4).
Migration_n_Query.ipynb	Neon PostgreSQL migration and interactive query tool (Section 5).
vaccination_er.png	Exported entity-relationship diagram image.
app.py	Streamlit dashboard source code (Section 7).
requirements.txt	Python dependencies for the Streamlit app.
secrets.toml.example	Template for the Neon connection string (real credentials are kept in Streamlit Cloud's own secrets manager, never committed).
gitignore	Prevents accidental commits of real secrets.

8.2 Tech Stack

- **Python** — pandas, numpy
- **SQLite** — initial normalized database, built in Colab
- **PostgreSQL (Neon)** — live, always-on production database
- **SQLAlchemy / psycpg2** — database connectivity from Python
- **Streamlit + Plotly** — interactive dashboard and visualizations
- **Graphviz** — entity-relationship diagram generation
- **Power BI** — Power BI Desktop connected via its native PostgreSQL connector

9. Consolidated Data Limitations

Rather than silently omit or guess at questions the data cannot answer, every limitation encountered across the EDA, SQL analysis, and both dashboards is stated explicitly at the point it arises, and summarized here:

- **No demographic breakdown** — No gender, education level, or within-country socioeconomic/income data (the closest available proxy is a cross-country World Bank income-group rollup, which answers a different question).
- **No sub-national geographic granularity** — No urban/rural split, no population density.
- **No sub-annual granularity** — All figures are annual; no month/date field exists to examine seasonal patterns.
- **No delivery-strategy field** — No way to compare door-to-door vs. centralized-clinic vaccination approaches.
- **No influenza case/incidence data** — Vaccine introduction status for seasonal influenza is tracked, but no influenza case counts exist anywhere in the disease-level tables.
- **Antigen-disease mapping is reference knowledge, not derived data** — `dim_antigen_disease_map` and `dim_intro_disease_map` are built from public WHO/CDC knowledge, not inferred from the data — not every antigen or disease in the dataset has a mapped counterpart.

10. Conclusion

This project took WHO vaccination data from raw, inconsistent CSV extracts all the way through to two live, independently-built interactive dashboards. Along the way, that meant cleaning and validating five separate datasets, designing a normalized SQL database with a proper star schema and documented ER diagram, answering every analysis question in the project brief with real SQL queries — stating explicitly where the data could not answer a question rather than guessing — migrating that database to a live PostgreSQL instance so it could be queried from anywhere, and building two different front ends on top of that same live data: a Power BI dashboard for a traditional business-intelligence experience, and a Streamlit application for a code-first, always-live experience with a manual refresh control.

The headline finding, consistent across the EDA, the SQL correlation analysis, and both dashboards, is that higher vaccination coverage is associated with materially lower disease incidence and case counts — while also surfacing exactly where and how vaccination programs are underperforming: specific low-coverage countries, regional disparities in both coverage and vaccine adoption, and a dose-retention problem distinct from initial vaccine access. These are the same findings a public health agency, vaccine manufacturer, or policymaker would need to prioritize interventions and allocate resources with evidence rather than guesswork.

References

- WHO Immunization Data Portal — source of all five raw datasets (vaccination coverage, incidence rate, reported cases, vaccine introduction, vaccination schedules).
- Neon (neon.tech) — PostgreSQL hosting for the live production database.
- Streamlit Community Cloud — hosting for the live dashboard.